ABESIT
Group of Institutions

# PitchX Abesit

- **Problem Statement Title - A Smart Platform for Predicting Health Risks, Managing Emergencies, and Supporting Patient Recovery**
- **Theme - Healthcare and Wellness**
- **PS Category - Software**
- **Team Name – UNFAZED X**
- **Team Members – Pragya Snehi , Prachi Kumari , Manish , Kunal**

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# Rakshak AI :
# Your 360° health emergency shield

## Problem Statement

- Health emergencies often occur **without early monitoring or prediction.**
- **Medical history is not instantly accessible** to hospitals during emergencies.
- **Family members may not receive immediate alerts** when a person is in danger.
- **Finding blood donors, hospitals, or medicines quickly** during emergencies is difficult.
- **Lack of preventive health monitoring** leads to late diagnosis of serious diseases.




## Innovation of the Solution.

Key innovative features:

- **AI Health Risk Radar** – Predicts health risks using lifestyle data.
- **Emergency Medical Passport (QR)** – Instant access to medical records.
- **Smart SOS System** – Sends alerts with live location to family.
- **Blood & Medicine Finder** – Finds nearby donors, hospitals, pharmacies.
- **Recovery Tracker** – Monitors patient recovery progress.

## Uniqueness

Unlike traditional systems that act **after emergencies**, our platform focuses on **early risk prediction, Smart SOS System, Emergency Medical Passport (QR), Blood & Medicine Finder and recovery monitoring.**

## Technical Highlight

**Real-time Biometric Sync:** Leverages Google Fit API to transform the AI Risk Radar from static analysis into a dynamic, 24/7 preventive monitoring system.

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# TECHNICAL APPROACH

## Technologies to be Used

### Frontend

**React.js** (Vite) – Fast and modern UI development
**Framer Motion** – Smooth animations and transitions
**React Router DOM** – Client-side routing
**Axios –** API communication with backend
**Leaflet.js** – Map integration (OpenStreetMap)
**Chart.js / Recharts** – Data visualization for recovery tracking

**Backend**

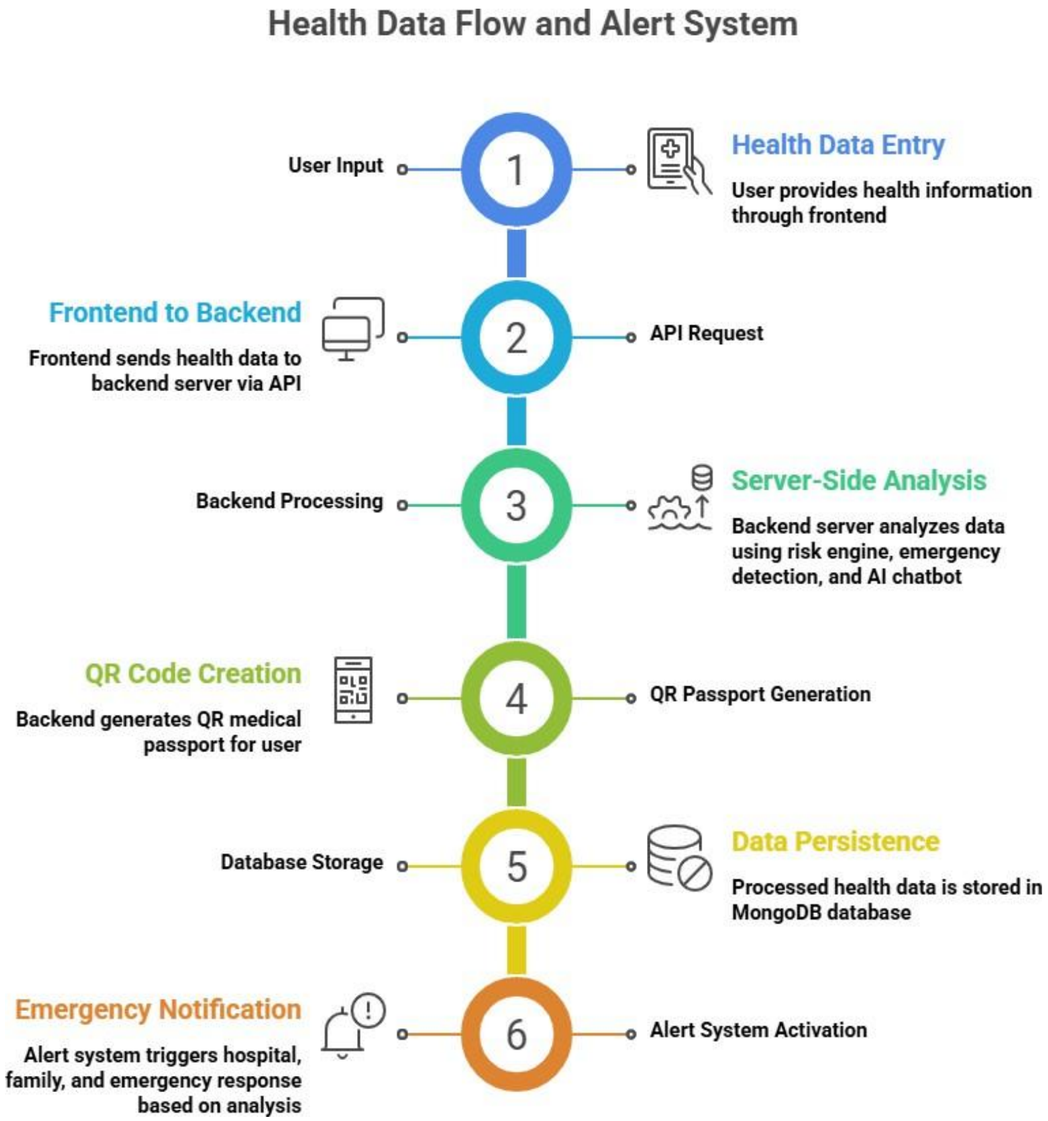
### Backend

**Node.js, Express.js** – Server-side logic and API development
**QRCode (NPM) -** Scannable Medical Passport system
**Browser Geolocation -**Native GPS for emergencies.
**Axios-** Streamlined Client-Server communication

### Deployment & Tools

**Vercel / Netlify** – Frontend deployment
**Render / Railway** – Backend hosting
**GitHub, VS Code** – Development and version control
**QR Code Generator API** – Emergency Medical Passport generation

## Methodology and Process for Implementation



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# ARCHITECTURE / FLOW DIAGRAM

## Functional Pipeline (Internal System Logic)

### Input Layer

User data collected from:
Health inputs
SOS trigger
Location services

⬇

### Processing Layer

Backend processes data for:
Health risk calculation
Emergency alerts
Resource search (donors, hospitals, pharmacies)

⬇

### Data Layer

Database stores:
User profiles
Medical history
Emergency logs
Recovery records

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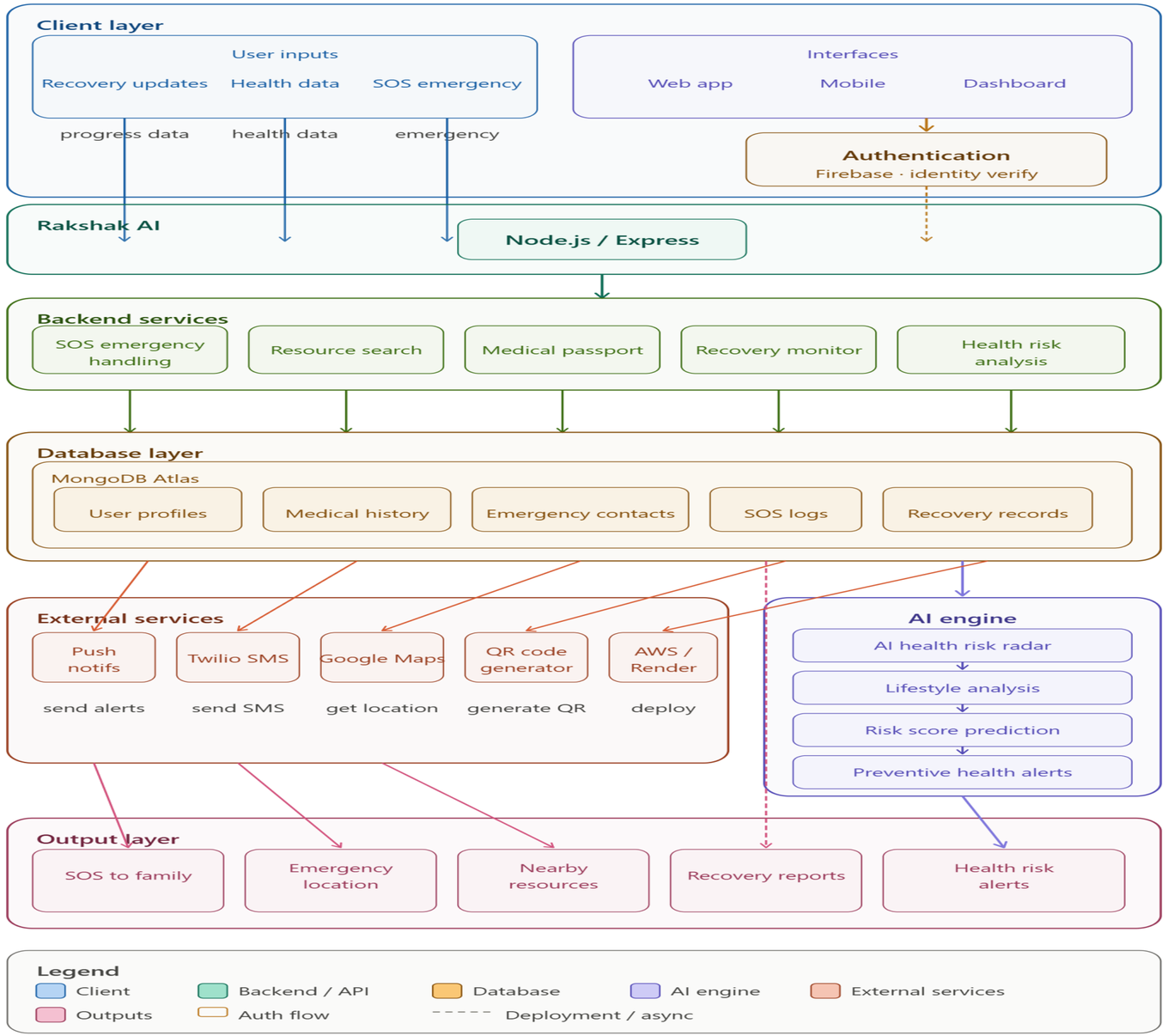
### Integration Layer

External services used for:
Location tracking
SMS alerts
QR code generation

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### Output Layer

System provides:
Health risk alerts
Emergency notifications
Recovery reports
Nearby medical resources

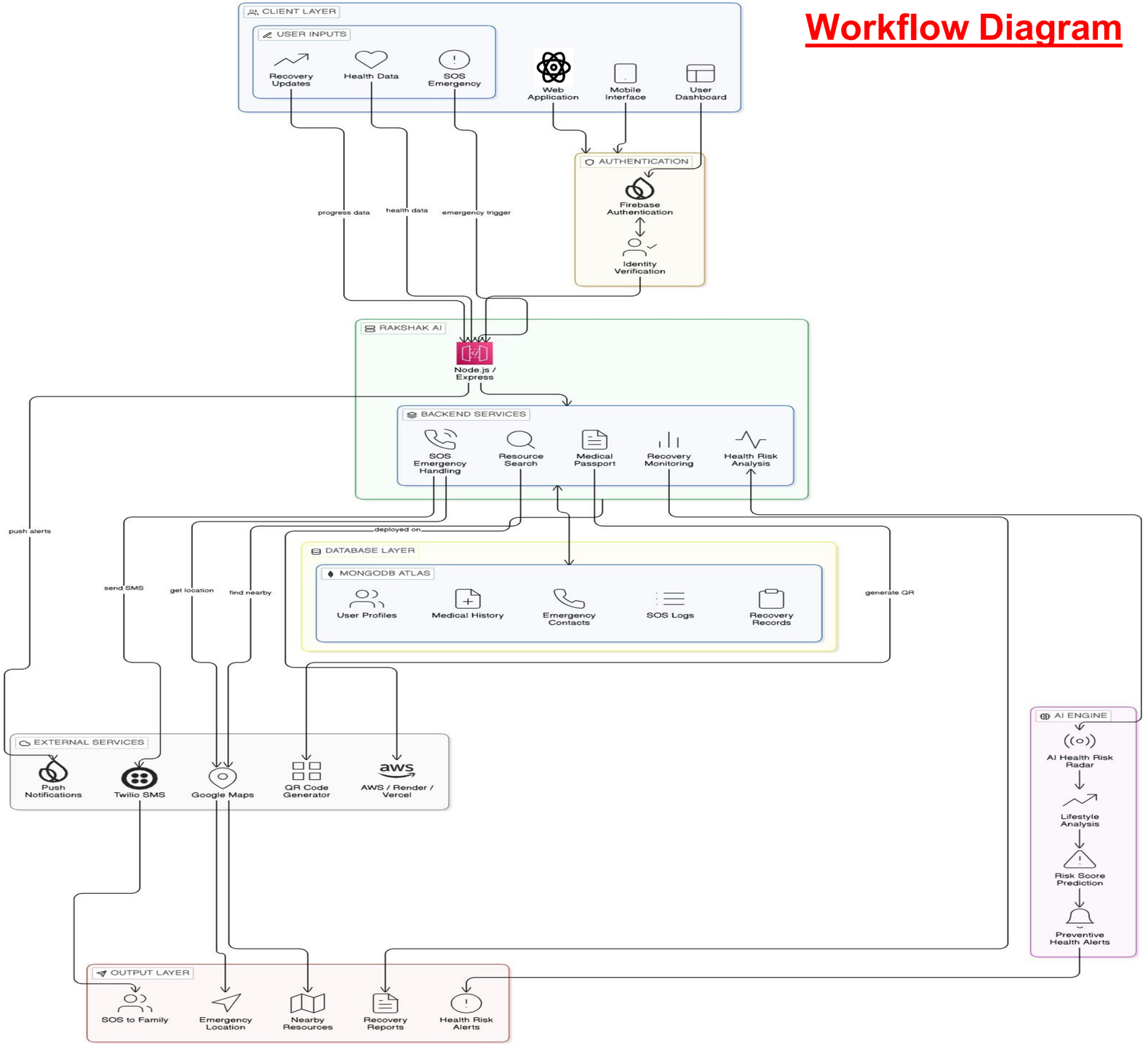


# Rakshak AI

Your 360° Health Emergency Shield




## Workflow Diagram



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# FEASIBILITY AND VIABILITY

## Feasibility & Viability Analysis

• **Technical Feasibility** – The system can be built using widely used web technologies such as **Angular**, **Node.js**, and **MongoDB**, making **development practical and scalable.**

• **Operational Feasibility** – The platform only requires a smartphone or internet-enabled device, making it **accessible to a large population.**

• **Economic Feasibility** – Most tools and frameworks used are **open-source, reducing development cost.**

• **Scalability** – Cloud-based architecture allows the system to support **large numbers of users in the future**.

## Business Potential

•Growing demand for **digital healthcare and emergency response platforms**.
•Can partner with **hospitals, clinics, and healthcare providers** for service integration.
•Opportunities for **subscription-based premium health monitoring services**.
•Potential collaboration with **insurance companies and telemedicine platforms**.
•Scalable model that can expand to **smart cities and global healthcare markets**.

## Potential Challenges and Risks

• **Accuracy of health risk prediction** if user inputs incorrect lifestyle data.
• **Data privacy and security concerns** related to sensitive medical information.
• **Dependence on external APIs** such as location or messaging services.
• **Availability of blood donors or medicines** in emergency situations.

## Strategies to Overcome Challenges

• Implement **data validation and user guidance** to improve input accuracy.
• Use **secure authentication and data encryption** to protect medical information.
• Integrate **multiple API services** to avoid dependency on a single provider.
• Build a **larger donor and hospital network database** to improve emergency resource availability.

## Supporting Facts for Feasibility & Viability

•**WHO:** ~19.8M annual deaths from cardiovascular diseases → need for early monitoring.
•**PubMed studies:** Lifestyle monitoring helps prevent many non-communicable diseases.
•**World Heart Federation:** 75% of CVD deaths occur in low- and middle-income countries.
•**McKinsey:** AI-driven digital health platforms are rapidly expanding in healthcare.

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# IMPACT AND BENEFITS

## Target Audience:

- **Chronic disease** patients
- Elderly individuals
- Working professionals & students
- Families and caregivers
- Hospitals and healthcare providers
- Blood donors and volunteers

## Potential Impact on Target Audience

- **Early detection** of health risks
- Faster emergency response with **alerts & live location**
- Instant access to **medical history via QR Passport**
- **Quick search** for nearby blood donors, hospitals, and pharmacies
- Better **recovery tracking** after treatment

## Benefits of the Solution

### Social Benefits

- Improves **public health awareness** and promotes preventive healthcare.
- Strengthens support between patients, families, and healthcare providers.

### Economic Benefits

- **Reduces medical costs** by detecting risks early and preventing severe health complications.
- Saves time and resources during emergency situations.

### Healthcare Benefits

- Improves **emergency response efficiency.**
- Provides **better health monitoring and recovery management.**

### Technological Benefits

Uses **AI, cloud computing, and real-time APIs** for smart healthcare management.
Integrates **multiple healthcare services into a single digital platform**.

### Environmental Benefits

Reduces **paper-based medical records** through digital QR medical passports.
Supports **remote monitoring**, reducing unnecessary hospital visits and travel.

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# RESEARCH AND REFERENCES

## RESEARCH

- Cardiovascular diseases are the **leading cause of death globally**, with about **19.8 million deaths annually**. - **World Health Organization**
- Many heart diseases can be **prevented through lifestyle monitoring and early intervention**.- **Global Health Research**
- **Over 75% of cardiovascular deaths** occur in low- and middle-income countries. - **World Heart Federation**
- **Non-communicable diseases cause around 41 million deaths every year**, many preventable with early monitoring.

## REFERENCES


World Health Organization
https://www.who.int
Health Risk Prediction Research
https://pubmed.ncbi.nlm.nih.gov
Emergency Healthcare Systems
https://www.ncbi.nlm.nih.gov
AI in Healthcare Reports
https://www.mckinsey.com


## Comparison with Existing Systems

| Feature | Traditional Emergency Apps | Rakshak AI |
|---|---|---|
| **Health Risk Prediction** | ❌ Not available | AI-based lifestyle risk prediction |
| **SOS Emergency Alerts** | ✅ Basic SOS alerts | Smart SOS with live location |
| **Medical Information Access** | ⚠️ Limited or manual | QR-based Emergency Medical Passport |
| **Resource Search (Hospitals/Donors)** | ⚠️ Limited features | Blood, hospital, and pharmacy finder |
| **Recovery Monitoring** | ❌ Not supported | Recovery tracking after treatment |
| **Integrated Healthcare Platform** | Focus only on emergencies | Predict → Protect → Rescue → Recover |